

PictorialAttributes: Depicting Multiple Attributes with Realistic Imaging

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ABSTRACT

Traditional visualizations often use abstract graphics, limiting understanding and memorability. Existing methods for pictorial visualization are more engaging, but often create disjointed compositions. To address this, we propose *PictorialAttributes*, a technique utilizing LLMs and diffusion models to depict data attributes. Examples show its promise for compelling and informative pictorial visualizations.

CCS CONCEPTS

• Computing methodologies → Computer graphics.

KEYWORDS

Pictorial Visualization, Image Generation

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1 INTRODUCTION

Visualization has long been indispensable for conveying complex information in an accessible manner. However, current visualizations often rely on abstract graphics like circles or bars, which can hinder comprehensive understanding and memorability, especially for data with multiple semantic attributes. Recent studies highlight that human-recognized objects are more memorable than abstract representations [Borkin et al. 2013], suggesting that pictorial visualizations using real images could significantly enhance visual perception [Romat et al. 2020; Shi et al. 2022]. Despite their potential, traditional methods of integrating realistic patches often rely on cut-and-paste techniques, which may result in disjointed compositions [Zhang et al. 2020]. Therefore, there is a growing

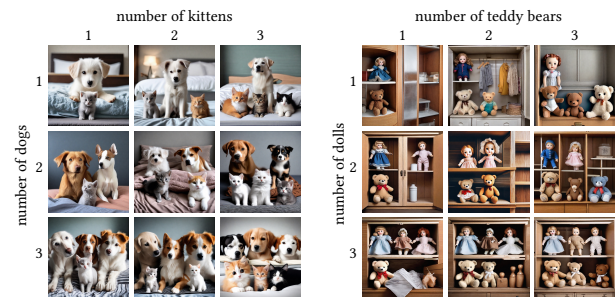


Figure 1: Our method using BA [Dahary et al. 2024] with SDXL [Podell et al. 2023] enables generating realistic images with specific quantitative attributes, e.g. object counts.

need for innovative approaches that facilitate seamless integration of pictorial elements into visualizations.

In response, we introduce *PictorialAttributes*, a novel technique for generating cohesive pictorial visualizations that can encode multiple data attribute. Unlike conventional methods relying on manual image placement [Xia et al. 2018], *PictorialAttributes* leverages large language models (LLMs) [Ouyang et al. 2022] and text-to-image diffusion models [Podell et al. 2023; Rombach et al. 2022] in a two-stage methodology. First, we employ LLMs to plan cohesive layouts tailored to varying quantitative attributes. Then, utilizing recent advancements like Bounded Attention (BA) [Dahary et al. 2024] and MIGC [Zhou et al. 2024], we generate realistic images according to these layouts. Through several examples, we demonstrate our method's effectiveness in accurately depicting multiple attributes, offering both informative and compelling visualizations.

2 METHOD

Our method takes the input of a family of parametric prompts y describing multiple objects and their attributes. Formally, we define y as a function $y : \mathcal{A} \rightarrow \mathcal{Y}$, where $\mathcal{A}$ is the set of all possible attribute combinations, and $\mathcal{Y}$ is the set of non-parametric textual prompts. These attributes can encompass both numeric and textual descriptors. For instance, y could represent prompts like "a cake with α layers and β candles", where $\mathcal{A}$ consists of ordered pairs of positive integers denoting the number of cake layers and candles. Similarly, employing textual attributes, y might take

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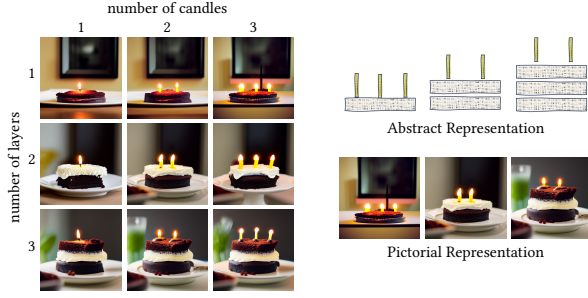


Figure 2: Our pictorial results compared to abstract visuals.

the form of "a lunch box with a α_1 amount of β_1 and α_2 amount of β_2 , ...", where $\mathcal{A}$ encompasses all possible series of pairs describing portion sizes $\alpha_i \in \{\text{small, medium, large}\}$ and ingredients $\beta_i \in \{\text{chips, noodles, rice, ...}\}$.

To generate various image variations of the input parametric prompts, our method operates in two stages. Initially, we plan a coarse layout for the generated image, effectively conveying the desired attributes. Subsequently, this formed layout, in conjunction with the input prompt, is fed into a layout-to-image model to produce an image aligning with the layout. This methodology ensures the requisite attributes are accurately depicted in the final image.

Layout formation. In this stage, we construct a parametric layout comprising parametric bounding boxes $\{b_i(a)\}_{i=1}^{n(a)}$ and local prompts $\{y_i(a)\}_{i=1}^{n(a)}$, where $n(a)$ denotes the number of objects in a specific realization of the parametric prompt $y(a)$, $a \in \mathcal{A}$.

To automate this process, we utilize LLMs, which were shown to excel in inferring layouts from prompts due to their understanding of real-world concepts and spatial reasoning [Lian et al. 2023; Qu et al. 2023]. For example, when generating layouts for Figure 2, a LLM comprehends that the cake layers should be stacked atop each other, while the candles should be placed on top.

Specifically, we input ChatGPT [Ouyang et al. 2022] with our parametric prompt y , enumerating all possible attribute choices in $\mathcal{A}$. We then prompt the model to generate a concise Python function. This function receives an attribute realization $a \in \mathcal{A}$ and outputs suitable bounding boxes $\{b_i(a)\}_{i=1}^{n(a)}$ and local prompts $\{y_i(a)\}_{i=1}^{n(a)}$, facilitating the generation of layouts dynamically.

Image generation. To generate the images, we employ diffusion text-to-image models, renowned for their ability to produce realistic high-quality images across a wide variety of domains. Specifically, we utilize the openly available Stable Diffusion [Rombach et al. 2022] (SD) and SDXL [Podell et al. 2023]. To further condition the image on the layout from the previous stage, we employ two recent state-of-the-art layout-to-image techniques tailored for generating multiple objects: Bounded Attention [Dahary et al. 2024] for SDXL and MIGC [Zhou et al. 2024] for SD.

3 RESULTS

Object counts. As shown in Figure 1 and on the left of Figure 2, our technique can fuse varying numbers of objects in a realistic and cohesive manner.



Figure 3: Visualization of dietary lunch boxes over six days.

Diet visualization. Figure 3 demonstrates our technique for visualizing a six-day food diet, showing diverse portion sizes and ingredient combinations in each lunch box. These visualizations clearly depict the dietary plan; for example, comparing day 1 and day 4 lunch boxes reveals differences in ingredient quantities.

4 FUTURE WORK

In this study, we introduce *PictorialAttributes*, a method using LLMs and diffusion models to create cohesive visualizations that encode multiple attributes. Initially designed for discrete data, extending our method to support high-volume and complex data types, including continuous values, is essential. However, our current approach is constrained by the limited capabilities of LLMs in layout planning and the diffusion model's accuracy in following layouts. Overcoming these challenges will be crucial for advancing future research.

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